

Date	30 June 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Yes	Proper indentation used in Python, HTML, and CSS files
2	Proper File Structure	Yes	Files organized into templates, static, and backend files
3	Meaningful Variable Names	Yes	Variables like N, P, K, humidity, rainfall are understandable
4	Function / Method Names	Yes	Functions like home(), prediction(), predict() are clear
5	Code Comments	Partial	Basic comments added, can improve with more explanations
6	Modular Design	Yes	Backend, frontend, and ML model separated properly
7	No Redundant Code	Yes	No unnecessary duplicate code
8	Error Handling	Yes	Try -except used in prediction route

Reusable Components / Modules:

S.No	Component / Module Name	language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Backend	Python	Request handling	High
2	Prediction Model	Machine Learning	Crop prediction	High
3	HTML Templates	HTMI/CSS	Fronted pages	High
4	Static Files	CSS/Media	UI design	Medium
5	Scaler Module	Scikit-learn	Input preprocessing	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized project structure
Readability	4	Easy to understand
Reusability	5	Components reusable in future
Documentation / Comments	4	Can improve with more comments
Overall Score	4.7	Good quality project code